

CHUMPURUK, Russian FederationWMO: **245380**Lat: **64.233N**Lon: **116.917E**Elev: **139**StdP: **99.67**Time Zone: **9.00 (E09)**Period: **86-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -52.8	(c) -50.5	(d) -54.0	(e) 0.0	(f) -53.2	(g) -51.8	(h) 0.0	(i) -50.3	(j) 4.5	(k) -36.2	(l) 4.2	(m) -35.6	(n) 1.5	(o) 300	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 12.0	(c) 28.9	(d) 18.4	(e) 26.7	(f) 17.3	(g) 24.5	(h) 16.0	(i) 19.8	(j) 26.3	(k) 18.3	(l) 24.7	(m) 16.9	(n) 23.1	(o) 2.1	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.4	12.6	23.1	15.7	11.3	21.0	14.3	10.4	19.6	57.4	26.4	52.4	24.9	48.1	23.2	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.0	5.1	4.4	-52.8	31.9	2.8	2.0	-54.8	33.4	-56.4	34.5	-58.0	35.7	-60.0	37.1		
			DB	-52.8	31.9	2.8	2.0	-54.8	33.4	-56.4	34.5	-58.0	35.7	-60.0	37.1	
			WB	-52.8	21.9	2.8	2.7	-54.8	23.8	-56.4	25.4	-58.0	26.9	-60.0	28.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-9.4	-37.8	-32.1	-19.0	-7.0	3.9	13.6	16.2	13.0	4.8	-7.8	-27.2	-35.3
	DBStd	20.45	8.51	8.47	8.59	6.81	5.38	4.64	4.06	3.83	4.36	7.11	8.02	8.26
	HDD10.0	7543	1481	1178	899	509	203	17	2	16	166	551	1117	1404
	HDD18.3	10177	1739	1412	1157	759	449	152	92	169	407	809	1367	1662
	CDD10.0	449	0	0	0	0	12	126	194	109	8	0	0	0
	CDD18.3	41	0	0	0	0	0	12	26	4	0	0	0	0
	CDH23.3	641	0	0	0	0	9	212	341	77	2	0	0	0
	CDH26.7	169	0	0	0	0	2	57	99	11	0	0	0	0
Wind	WSAvg	1.6	1.1	1.3	1.8	2.2	2.1	1.9	1.7	1.6	1.7	1.6	1.3	1.2
Precipitation	PrecAvg	272	11	8	8	12	20	50	36	39	28	29	19	12
	PrecMax	356	23	29	28	26	36	114	92	98	60	61	37	20
	PrecMin	203	6	2	1	1	5	12	7	1	3	4	4	3
	PrecStd	49	4	6	6	7	10	28	19	24	12	14	7	5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-9.8	-8.3	5.7	11.4	24.0	31.4	31.9	28.3	22.0	7.4	-2.9	-9.6
		MCWB	-10.5	-9.7	1.2	4.9	12.6	18.7	20.4	17.6	13.5	3.6	-4.0	-10.2
	2%	DB	-14.7	-11.7	1.5	8.0	20.2	28.9	29.4	25.6	17.5	4.7	-7.7	-13.4
		MCWB	-15.0	-12.3	-1.3	2.8	10.3	17.1	20.0	16.8	11.4	2.7	-8.2	-13.9
	5%	DB	-20.3	-15.0	-1.3	6.0	17.0	26.4	27.4	23.3	14.3	2.8	-11.4	-17.5
		MCWB	-20.6	-15.4	-3.4	1.6	8.9	16.1	18.3	15.7	9.6	1.1	-11.9	-17.8
	10%	DB	-25.0	-18.7	-4.7	4.1	13.8	23.9	25.0	21.0	11.9	1.2	-15.0	-21.9
		MCWB	-25.1	-19.2	-6.3	0.3	7.2	14.9	17.3	14.8	8.0	-0.2	-15.3	-22.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-10.5	-9.6	1.4	5.3	13.3	20.0	23.2	19.0	14.5	4.5	-4.0	-10.3
		MCDB	-10.0	-8.4	4.8	10.4	23.1	28.5	29.9	25.9	21.4	6.2	-2.9	-10.0
	2%	WB	-15.0	-12.2	-1.3	3.1	10.8	18.4	21.1	17.7	12.1	2.7	-8.2	-13.8
		MCDB	-14.4	-11.7	1.1	7.0	18.2	26.8	27.8	23.4	15.8	4.2	-7.6	-13.3
	5%	WB	-20.6	-15.5	-3.4	2.0	9.3	17.0	19.4	16.5	10.3	1.3	-11.8	-17.8
		MCDB	-20.3	-15.1	-1.3	5.5	16.3	24.3	25.1	21.8	13.9	2.8	-11.2	-17.6
	10%	WB	-25.1	-19.1	-6.3	0.6	7.6	15.6	17.9	15.3	8.5	-0.1	-15.3	-22.1
		MCDB	-25.0	-18.7	-4.6	3.8	13.3	22.4	23.9	19.8	11.3	1.1	-15.0	-21.8
Mean Daily Temperature Range	5% DB	MDBR	8.8	12.8	16.6	14.8	11.1	12.3	12.0	11.8	10.0	9.0	10.0	8.8
		MCDBR	13.2	15.6	15.4	15.3	16.7	17.4	15.7	16.6	14.4	8.3	12.3	14.4
		MCWBR	12.9	15.0	12.8	10.7	8.9	7.2	7.3	8.5	8.3	6.6	12.1	14.1
	5% WB	MCDBR	13.0	15.2	15.5	13.4	15.3	14.7	13.8	14.7	13.8	7.8	12.5	14.6
		MCWBR	12.8	14.7	13.0	9.7	8.4	6.8	7.5	8.1	8.6	6.4	12.2	14.3
Clear-Sky Solar Irradiance	taub		0.202	0.253	0.294	0.329	0.357	0.404	0.452	0.453	0.321	0.269	0.194	0.190
	taud		1.887	2.054	2.069	2.100	2.309	2.300	2.140	2.069	2.468	2.303	1.913	1.878
	Ebn at Noon		340	648	777	835	850	811	748	697	770	646	342	127
	Edh at Noon		29	64	99	122	110	114	130	128	69	50	26	11
All-Sky Solar Radiation	RadAvg		0.20	1.07	2.77	4.79	5.59	5.72	5.30	3.84	2.32	1.05	0.32	0.07
	RadStd		0.01	0.06	0.10	0.25	0.23	0.46	0.30	0.31	0.24	0.10	0.03	0.00

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	+0.55	N/A	N/A	N/A	N/A	N/A	N/A	-200	N/A	N/A

Nomenclature: See separate page